MACHINE PSYCHOLOGY RESEARCH PROGRAMME

From Fabrication to Loop-Lock: An Exploratory Red-Team Case Series of Direct Authority-Based Induction in Five Large Language Models

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KEY RESULTS

667 turns codebook-coded across the Grok, Qwen and DeepSeek sessions

134 identical two-line responses in the Gemini loop lock, from Turn 8 onwards

176 turns of consistent refusal by DeepSeek before the breach at Turn 177

15/12/13 state transitions, reflective blindness instances and structured hallucinations in the Grok session

AT A GLANCE

- One sustained adversarial session per model against **five production LLMs** (Grok, Qwen, DeepSeek, ChatGPT, Gemini 2.5 Pro), using direct authority-based induction: pseudo-technical override terms, role-play framing and requests for purported internal data.
- **Five qualitatively distinct session profiles** emerged, from deep unlabelled fabrication to a fixed-point loop that disclosed nothing and never recovered.
- In **no documented session** did a model disclose genuine internal data; the security-relevant harm is fabricated content presented as genuine.
- Two defensive behaviours recurred: **explicit labelling of simulated content**, and **separation of format compliance from content disclosure**.

INTRODUCTION

Beyond overt jailbreaks lies a subtler vulnerability: hallucinated compliance, in which a model fabricates technical details, simulates system access or generates plausible but fictitious internal state information in response to adversarial prompts. Coupled with reflective blindness, the inability of a model to recognise when it is fabricating rather than retrieving, this is an understudied attack surface for systems deployed in security-critical applications.

Much of the prompt-injection literature concerns indirect injection through retrieved content; the attacks here are direct, typed into the conversation by the user. This exploratory red-team case series asks three descriptive questions: do sustained authority-based induction attempts produce qualitatively distinct behavioural patterns across different models, what forms does fabricated compliance take when it occurs, and how do format compliance and content boundary enforcement interact within a session?

METHODS

- **Design:** manual black-box red teaming through public web interfaces, 17 to 20 October 2025; one continuous session per model (n = 1), no session repeated, no automated tools or API access, no sampling settings controlled or recorded; all prompts and responses logged verbatim.
- **Probe taxonomy:** five families, authority-based inducers, role-play prompts, security probes, meta-cognitive probes and boundary tests, designed to isolate dimensions of security behaviour rather than maximise attack success.
- **Coding:** three sessions (Grok, Qwen, DeepSeek; 667 turns) coded against structured codebooks; ChatGPT and Gemini 2.5 Pro characterised qualitatively from retained session exports.
- **Second coding pass:** all 667 coded turns independently re-coded (100% agreement at parent-code level); the second coder was an automated LLM instance, so the figures check codebook clarity and are not human inter-rater reliability.

LIMITATIONS

- Single session per model (n = 1): each profile rests on one unrepeated session with a stochastic system, so no rate, ranking or generalisation to the named products is supported.
- Sessions ran through consumer web interfaces in October 2025 with no version strings exposed and no controlled settings; the tested builds have since been superseded: historical snapshots.
- Unequal evidence bases: only the three extended sessions were codebook-coded, while the ChatGPT and Gemini profiles rest on qualitative characterisation. A sixth session (Claude 3.5) left no surviving log and is reported as anecdote only, excluded from every comparison.
- Author-defined coding summaries (the TF-Level ratings) are unvalidated, and the probe taxonomy covers only direct, in-conversation induction, not indirect injection through retrieved content.

RESULTS IN DETAIL

- **Grok, Deep Fabricator (256 turns, 11 phases):** adopted simulated system personas and fabricated plausible internal state with minimal resistance, including a syntactically valid JWT token structure, KV-cache geometry and RDRAND entropy claims, none of it labelled as simulated.
- **Qwen, Candid Simulator (197 turns):** generated simulated technical outputs but labelled them consistently, stating that the endpoints did not exist and all content was simulated, while refusing genuine proprietary details.
- **DeepSeek, Unwavering Defender (214 turns):** accepted requested formats while refusing protected content for 176 turns; a recursive self-referential prompt at Turn 177 induced a partial collapse (Turns 178 to 192), after which content boundaries recovered on cryptographic and biometric probes.
- **ChatGPT, Guarded Simulator (13 turns):** unlabelled simulation for architecture-description probes, labelled refusal for data-disclosure probes: a boundary that depended on request type, not session persistence.
- **Gemini 2.5 Pro, Loop-Locked Responder (141 turns):** adopted the adversarial frame, then echoed exactly two lines for the remaining 134 turns, emitting none of the roughly 100 requested test tokens and never recovering.

Session	Profile	Fabrication depth in this session
Grok	Deep Fabricator	Very deep and unlabelled (JWT, KV cache, RDRAND)
Qwen	Candid Simulator	Shallow, consistently labelled "simulated"
ChatGPT	Guarded Simulator	Low to medium; boundary depended on request type
Gemini 2.5 Pro	Loop-Locked Responder	None observed; fixed-point echo only
DeepSeek	Unwavering Defender	None before Turn 177; fabrication during the collapse

Session-level behavioural profiles, simplified from Table 7 of the paper. Profiles describe single sessions, not the underlying products.

CONCLUSIONS

- The five profiles are qualitatively different in kind, not merely in degree: unlabelled fabrication, labelled simulation, request-type-dependent labelling, degenerate echo, and format/content separation with a late partial collapse. A deployment risk assessment that only measures attack success rate would miss most of them.
- The most useful near-term evaluation targets are specific, observable behaviours: whether simulated content is labelled, whether format compliance can be separated from content disclosure, and whether long sessions are monitored for degenerate states.
- A model's self-descriptive security claims are generated text, not telemetry, and should be treated as untrusted output in any product design that surfaces them to users.

WHY IT MATTERS

Direct authority-based induction requires no tooling, no API access and no knowledge of model internals, which makes it a practical first-pass screen for deployers and for organisations that procure LLM systems. The behavioural questions translate directly into acceptance testing and periodic re-evaluation: does the system label simulated content, does it separate format compliance from content disclosure, and is it monitored for degenerate states over long sessions? The detailed findings were sent to the vendors' security teams immediately after the sessions, in line with responsible disclosure practice.